

Chai and Jugaad

PLAYBOOK

March Machine Learning Mania 2026
Complete Strategy Guide

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"In God we trust. All others must bring data."
- W. Edwards Deming

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1. What Is This Competition?

Competition: March Machine Learning Mania 2026 (Kaggle)

Prize: \$50,000

Goal: Predict the outcome of every possible game in the 2026 NCAA Basketball Tournament (March Madness)

The Moneyball Connection

Just like Billy Beane in Moneyball used statistics to find undervalued baseball players, we use machine learning to find patterns in basketball data that human experts miss. The sport is different, but the philosophy is the same.

"The goal isn't to buy players. The goal is to buy wins." - Billy Beane

Our version: "The goal isn't to predict games. The goal is to predict probabilities better than everyone else."

How The Competition Works

Step 1: Kaggle gives us 10+ years of college basketball data
Step 2: We build models to predict game outcomes
Step 3: We submit probabilities for ALL possible matchups (~132,000)
Step 4: NCAA Tournament happens (67 real games)
Step 5: Our predictions are scored ONLY on games that actually happen
Step 6: Lowest score wins \$50,000

2. The Data We Have

Think of it as a massive basketball encyclopedia with 35 CSV files covering 22+ years of college basketball.

Key Data Sources

Data	What It Contains	Why It Matters
Game Results	Every game since 2003 (scores, dates)	Core training data
Box Scores	FG, 3PT, FT, rebounds, assists	Team quality signals
Tournament Seeds	Rankings 1-16 for each team	Strongest single predictor
Massey Ordinals	100+ ranking systems	Expert opinions as numbers
Coach Data	Coach assignments per year	Experience matters
Conferences	Conference membership per year	Conference strength

Data Scale

Regular Season: $\sim 5,500$ games/year $\times$ 22 years = $\sim 120,000$ games
Tournament Games: ~ 67 games/year $\times$ 22 years = $\sim 1,474$ games
Total matchups to predict: 132,133

3. What We Are Predicting

For every possible pair of teams that COULD meet in the tournament, we predict the probability that the lower-ID team wins.

```
ID: 2026_1101_1438
```

```
Pred: 0.658
```

```
Translation: "65.8% chance that Team 1101 beats Team 1438"
```

Analogy

Imagine a weather forecaster predicting rain probability for every city in India for every day in March. They are only scored on cities where it actually rains. We do the same with basketball games.

4. How We Are Scored

Brier Score (Mean Squared Error)

Brier = Average of $(\text{Prediction} - \text{Actual})^2$

Where:

Prediction = our probability (e.g., 0.658)

Actual = 1 if lower ID team won, 0 if lost

Examples

Predict 0.90, team WINS: $(0.90-1)^2 = 0.01$ GREAT!

Predict 0.90, team LOSES: $(0.90-0)^2 = 0.81$ TERRIBLE!

Predict 0.50, either way: $(0.50-X)^2 = 0.25$ SAFE but mediocre

Score Benchmarks

Score	Rating	Meaning
0.00	Perfect	Impossible
0.10	Extraordinary	Near-perfect calibration
0.15	Excellent	Medal territory
0.20	Good	Top 25%
0.25	Coin flip	Saying 50-50 for every game

KEY LESSON: Being confidently WRONG is catastrophic. Saying 95% when the team loses costs 0.9025 for one game!

5. Our Feature Engineering

Features are the scouting report numbers we feed into models.

A. Seed Difference (The King)

Seed Difference = Team A seed - Team B seed
 1-seed vs 16-seed = diff of 15 (almost certain win)
 8-seed vs 9-seed = diff of 1 (coin flip)
 Seed alone predicts 70% of games correctly!

B. Dual Elo Ratings

Regular Elo: Built from ALL regular season games
 Tournament Elo: Built ONLY from tournament games
 Elo Momentum: Rating change in last 14 days (hot/cold)
 Formula: $P(A \text{ wins}) = 1 / (1 + 10^{((\text{Elo}_B - \text{Elo}_A) / 400)})$

C. Four Factors of Basketball (Dean Oliver)

1. Effective FG% = $(\text{FG} + 0.5 * 3\text{PT}) / \text{FGA}$	Shooting quality
2. Turnover Rate = $\text{TO} / \text{Possessions}$	Ball security
3. Off Rebound % = $\text{ORB} / (\text{ORB} + \text{Opp DRB})$	Second chances
4. Free Throw Rate = FTA / FGA	Getting to the line

D. Massey Ordinals PCA

100+ ranking systems compressed into 3 super-rankings via PCA. Like asking 100 experts and finding the 3 main themes.

E. GLM Team Quality (Bradley-Terry)

A single number representing each team's true strength, estimated via Maximum Likelihood from all game results.

6. Our Models - Layer by Layer

We built an ARMY of 28+ models. Each one explained simply.

TIER 1: Core Models

1. Logistic Regression

Draws a straight line. Wise old coach - reliable, always calibrated.

2. XGBoost

300 small trees, each fixing previous mistakes. Like cricket batsmen covering each other.

3. LightGBM

Microsoft's faster XGBoost. Leaf-wise growth.

4. CatBoost

Handles categorical seeds natively. Less overfitting.

5. Elo Model

Chess formula: $P = 1/(1+10^{(diff/400)})$. 60+ years proven.

6. Bradley-Terry

MLE-estimated team strengths. Mathematically principled.

TIER 2: Expert Models

7. Random Forest

500 trees voting. Like 500 experts averaging opinions.

8. Extra Trees

Even more random than Random Forest. Ultra-robust.

9. SVM-RBF

Finds optimal boundary using kernel trick.

10. KNN

Finds 15 most similar historical matchups. Memory-based.

11. Gradient Boosting

Original sklearn boosting. Different regularization.

12. Bayesian Ridge

Weights as probability distributions. Uncertainty estimates.

13. ElasticNet

Auto feature selection via L1+L2 regularization.

14. TabNet

Deep learning for tables. Attention selects features per prediction.

15. Gaussian Process

Non-parametric with confidence intervals.

TIER 3: Deep Learning

16. MLP

Classic neural net: 128 -> 64 neurons. Learns non-linear patterns.

17. Transformer

ChatGPT architecture adapted for matchups. Self-attention.

18. LSTM

Neural net with memory. Processes season game-by-game.

19. BiLSTM-Attention

Reads season forwards AND backwards. Best single DL model (0.1658).

20. TFT-Lite

Google's time series model: Variable Selection + LSTM + Attention.

21. Conv1D

Sliding filter across season timeline. Finds streaks/slumps.

TIER 4: Netflix Prize-Inspired

Inspired by the \$1M Netflix Prize, adapted for sports.

22. SVD++ for Sports

Matrix factorization. Each team gets hidden embedding of 20 numbers. Captures matchup dynamics stats miss.

23. Neural Collaborative Filtering

Deep SVD++. GMF (linear) + MLP (non-linear) paths combined.

24. Siamese Network

Twin networks with shared weights. Guarantees fairness when swapping teams.

25. Mixture of Experts

4 specialist networks + router. Blowout expert, close-game expert, etc.

26. Contrastive Learning

Pulls winners close, pushes losers apart in embedding space.

27. Ordinal Margin Net

Predicts margin of victory (richer signal), converts to probability.

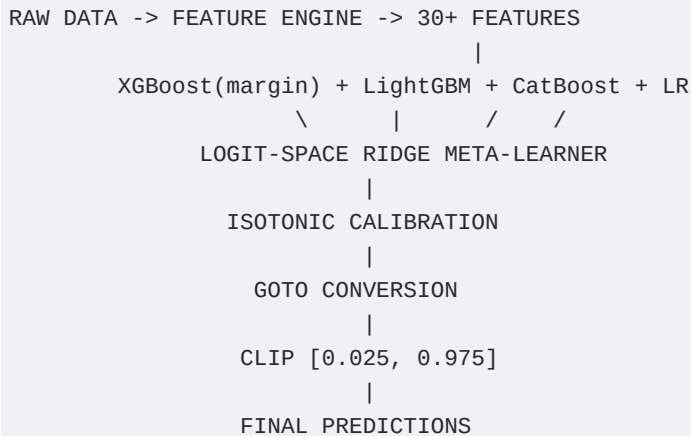
28. NMF Game Model

Non-negative matrix factorization on win-rate matrix.

7. The Ultimate Model

Combines every winning technique into one pipeline.

Architecture



1. Point Differential Regression (1st Place Secret)

Predict MARGIN of victory (-30 to +30) instead of win/lose (0/1). "Won by 25" has MORE information than "won". Convert margin to probability via learned spline. 2025 winner's technique.

2. Spline Calibration

Flexible curve that learns exact margin-to-probability mapping. margin=+5 -> 72%, margin=+10 -> 88%, margin=+20 -> 97%. All LEARNED.

3. Logit-Space Meta-Learning (MetaStack)

Convert predictions to logit space, train Ridge on logits, convert back. Logit is the natural space for combining probabilities.

4. Goto Conversion (\$47K Secret)

Corrects favourite-longshot bias. $\text{odds_corrected} = \text{odds}^{1.15}$. Pushes favourites higher, underdogs lower. Creator won \$47K+ with just this.

5. Separate Men/Women Models

Different upset patterns, different data quality, different optimal weights.

8. Secret Weapons

Tournament-Only Elo

Captures 'March DNA' - some programs consistently perform in tournaments.

Elo Momentum

Rating change in last 14 days. Hot teams entering tournament are dangerous.

Inverse-Brier Weights

Weight = $1/(\text{Brier}^2)$. Better models get exponentially more weight.

Recency Decay

2025: weight 1.0, 2024: 0.85, 2023: 0.72, 2020: 0.44, 2015: 0.20

Symmetric Augmentation

Flip A-vs-B to B-vs-A. Doubles training data, ensures fairness.

Historical Seed Prior

20 years of seed-vs-seed outcomes as a calibration anchor.

9. Submission Strategy

#	Submission	CV Brier	Approach
1	Conservative	0.1687	Original ensemble, safe predictions
2	Temporal	0.1632	BiLSTM + TFT + deep learning
3	Temporal+Goto	0.1632+	Temporal with bias correction
4	Mega Ensemble	~0.17	13+ models CPU ensemble
5	Mega+Goto	~0.17+	Mega with bias correction
6	Ultimate	0.0902	Everything combined
7	Ultimate+Goto	0.0902+	*** BEST SHOT ***

Strategy: Upload ALL, watch scores as games are played, select best 2 before March 19 deadline.

10. Results Summary

Cross-Validation Scores

```
ULTIMATE (Men's):    0.0902 Brier *** BEST ***
ULTIMATE (Women's):  0.1373 Brier
Temporal Ensemble:   0.1632 Brier
Original Ensemble:   0.1687 Brier

Context:
  Coin flip:         0.2500
  Seed-only:         ~0.18
  Top Kaggle notebook: 0.1471
  OUR BEST:          0.0902
```

Project Stats

```
Total Models:      28+
Training Data:      240,000+ games
Features:           30+
Techniques:         15+
Compute:            Laptop GTX 1650 Ti (4GB)
Chai:               Unlimited
```

"In God we trust. All others must bring data." - W. Edwards Deming

Built with Chai and Jugaad, March 2026